

Maryam Asif

B.S. Computer Science | FAST-NUCES

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TECHNICAL SKILL

Languages: C++, Python, JavaScript, SQL, HTML, CSS

Frameworks/Libraries: React (Basic), Streamlit, Scikit-learn

Tools: VS Code, Jupyter Notebook, VMware, Figma, DOSBox, Cisco Packet Tracer, XAMPP

PROJECTS

- **Multi-Campus Communication System (C++, TCP/UDP)** [Github](#)

Built a client-server system using C++ with TCP for reliable messaging and UDP for real-time broadcasting. Implemented multi-threading to handle multiple clients and track online status across campuses.

- **Hybrid AI Movie Recommendation System (Python, Streamlit, Scikit-learn)**

Built an AI-powered movie recommender using CSP, A* search, K-Means clustering, and ANN-based rating prediction with explainable recommendations and an interactive Streamlit UI.

- **Hybrid Core Banking System (Html, CSS, JS, PHP)** [Github](#)

Developed a centralized banking system to manage accounts, transactions, and fund transfers using relational database concepts. Applied normalization, SQL queries, and transaction handling to ensure data integrity and consistency

- **Google Drive Simulation (C++)** [Github](#)

Built a console-based cloud storage system using trees, hash tables, and graphs for efficient file management and sharing. Implemented file versioning, access control, compression, and user-based sharing. Focused on optimizing performance through data structure design

- **Dynamic Pathfinding Agent (Python, Tkinter)** [Github](#)

Developed a grid-based pathfinding visualizer using A* and Greedy Best-First Search with Manhattan and Euclidean heuristics. Implemented an interactive GUI with real-time visualization of search process, obstacles, and performance metrics.

- **Rock Paper Scissors Game (HTML, CSS, JavaScript)** [Github](#)

Developed a responsive web-based Rock Paper Scissors game with interactive UI, animations, and sound effects. Implemented game logic, random computer moves, and score tracking using JavaScript.

- **Women Safety App UI/UX (Figma)**

Designed a user-friendly mobile app interface in Figma focused on quick access to emergency features like SOS alerts, live location sharing, and trusted contacts. Created intuitive user flows and responsive layouts to enhance usability and accessibility.

- **Ping Pong Game (Assembly Language)**

Developed a console-based Ping Pong game in Assembly with real-time input handling and game logic.

Applied low-level programming concepts including registers, interrupts, and control flow for efficient execution.

Certificates & Participation

- ICIT Workshop 2024 – Participant
- Seminar: “Revolutionising Traditional Business Sectors”